

Contact Information

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Work experience

- **Associate Professor (with tenure)**, Hong Kong Baptist University, HK
2024.7-now, Department of Computer Science (early promotion)
- **Assistant Professor**, Hong Kong Baptist University, HK
2019.2-2024.6, Department of Computer Science
- **Honorary Research Officer**, Imperial College London, UK
2014.8-2017.7 Department of Medicine
- **Research Assistant**, The University of Hong Kong, HK
2008.10-2010.9 Department of Paediatrics and Adolescent Medicine

Education Background

- **Postdoctoral Fellow**, Stanford University, USA
2016.6-2019.1, Department of Computer Science and Department of Pathology
Supervisor: Prof. Serafim Batzoglou, Prof. Arend Sidow
- **Postdoctoral Scholar**, City University of Hong Kong, HK
2016.2-2016.5, Department of Computer Science
Supervisor: Prof. Shuaicheng Li
- **Visiting PhD student**, UC Berkeley, USA
2015.1-2015.7, Department of Mathematics
Supervisor: Prof. Stephen Smale
- **PhD**, City University of Hong Kong, HK
2012.10-2016.2, Department of Computer Science
Supervisor: Prof. Shuaicheng Li
- **MPhil**, The University of Hong Kong, HK
2010.9-2012.9 Department of Paediatrics and Adolescent Medicine
Supervisor: Prof. Wanling Yang
- **BEng**, Tianjin University, Tianjin, China
2004.9-2008.7 School of Software Engineering

Research Interests

- Multi-agent for Biomedical Research
- Develop deep learning models for single-cell sequencing and spatial multi-omics data analysis
- AI for genomics, microbiology and metabolomics
- AI for Traditional Chinese Medicine

Grant

- Empowering Traditional Medicine with Modern Technology: Application and Research of Large Language Models in TCM Education. Chinese Medicine Development Fund (**PI, 1,994,080 HKD**, Approved in principle)
- Biomedical AI Agent for Traditional Chinese Medicine Treatment of Gastrointestinal Cancer: Research on Microbiome-Host Interaction Mechanisms. Guangdong and Hong Kong Universities “1+1+1” Joint Research Collaboration Scheme (**PI, 1,500,000 HKD**)
- Enhancing Small Molecule Drug Synergy Prediction Using Multi-Modality and Multi-source Data. RCMS/24-25/03, the Research Cluster Matching Scheme (**PI, 1,000,000 HKD**, Internal grant, 2025.9-2027.8)
- Towards Trustworthy Foundation Models under Imperfect Scenarios. Young Collaborative Research Grant, RGC/C2005-24Y (**Co-PI, 4,006,800HKD; allocated 789,360 HKD**, 2025.7-2028.6)

- Improved Metabolic Network Reconstruction and Metabolite Profiles Prediction using Complete and Strain-Resolved Microbial Genomes. Young Collaborative Research Grant, RGC/C2004-23Y (**PC, 4,494,061HKD**, 2024.6-2027.5).
- Advancing clinical differentiation between constipation-predominant irritable bowel syndrome and functional constipation using machine learning models. Health and Medical Research Fund, 11221026 (**PI, 487,800HKD**, 2024.9-2027.8).
- Computational models for health management system using multi-source data, Guangdong-Hong Kong Technology Cooperation Funding Scheme, GHX/133/20SZ, (**PI, 1,844,600HKD**, 2022.9-2024.8).
- 基于长读长测序技术的宏基因组组装算法研究, BGI Open Fund, BGIRSZ20220014, (**PI, 450,000RMB**, 2023.1-2024.12).
- A computational framework to prioritize disease-associated low frequency variants from Identity-By-Descent regions, ECS HKBU22201419, (**PI, 651,769HKD**, 2019.9-2023.2)
- A novel metagenome assembly algorithm by using single-tube long-fragment sequencing. Guangdong Basic and Applied Basic Research Foundation 2019A1515011046, (**PI, 100,000RMB**. 2019.9-2022.8)
- A graph-based diploid assembly algorithm for detecting structural variants on 10x Linked-Read sequencing. Guangdong Basic and Applied Basic Research Foundation 2021A1515012226, (**PI, 100,000RMB**, 2021.9-2024.8)
- 一种基于单体型局部组装和共有单体型分析的复杂疾病罕见变异识别算法, Shenzhen Virtual University Park (SZVUP) Special Fund 2021Szvup135, (**PI, 187,000RMB**, 2021.8-2023.7).
- A High-resolution Global Metagenome Catalog from Uncultured Human Gut Microbiota. IRCMS/19-20/D02, Interdisciplinary Research Clusters Matching Scheme (**PI, 637,500HKD**, Internal grant)
- A computational network-based framework for analyzing single-cell sequencing. Startup grant Tier 2 RC-SGT2/19-20/SCI/007, (**PI, 480,000HKD**, Internal grant)

Awards and Honors

- Best Research Award, Department of Computer Science, Hong Kong Baptist University, 2025
- Excellent Paper Award, Faculty of Science, Hong Kong Baptist University, 2024
- President's Award for Outstanding Performance as Early Career Researcher, Hong Kong Baptist University, 2023
- Best Teaching Award, Department of Computer Science, Hong Kong Baptist University, 2022
- Travel Grants for Exchange Activities of Research Students, City University of Hong Kong, 2015
- University Postgraduate Fellowship, City University of Hong Kong, 2012-2015
- University Postgraduate Fellowship, The University of Hong Kong, 2010-2012
- Best practice award, Bio-IT World 2012
- Second roll of Imagine Cup 2007
- First-class prize in the China Adolescents Science and Technology Innovation Competition in 2003
- First-class prize in the China Adolescents Science and Technology Innovation Competition in 1998

Publications (†co-first author, *corresponding author)

Published Journal Papers

Under Review and Preprint

- Honghao Lai, Weilong Zhao, Hanxiang Liu, Jiayi Liu, Ning Ma, Bei Pan, Xufei Luo, Min Han, Danni Xia, Bo Tong, Ju Wang, Zheng Zhang, Jing Xiong, Rui Yang, Yiyun Wang,

Jinhui Tian, **Lu Zhang**, Janne Estill, Yinghui Jin, Hongcai Shang, Zhaoxiang Bian, Kehu Yang, Nannan Shi, Yaolong Chen, Long Ge*. ChatCPM: An Artificial Intelligence System for Evidence-Based Consultations on Chinese Patent Medicines. Under review in **Journal of Advanced Research (Q1, IF:13.45)**.

- Yi Ding, Jin Xiao, Bohao Zou, Chao Yang **Lu Zhang***. DeepMM: Identify and correct Metagenome Misassemblies with deep learning. Under Review in **Genome Research (Q1, IF: 5.5)** doi: 10.1101/2025.02.07.637187
- Bohao Zou[†], Zhenmiao Zhang[†], Xiaohan Wang, Rong Tao, Nianzhen Gu, Karsten Kristiansen, Mo Han*, **Lu Zhang***. CompleteBin: A transformer-based framework unlocks microbial dark matter through improved short contig binning. Major Revision in **Nature Biotechnology (Q1, IF: 41.7)**. doi:10.1101/2025.04.20.649691v3
- Chonghao Wang, **Lu Zhang***. A structure-informed evolutionary model for predicting viral immune escape and evolution. Major Revision in **Nature Computational Science (Q1, IF:18.3)**. doi: 10.1101/2025.07.31.667864
- Yingqiao Zhang, Zeming Li, Bo Han, Aiping Lyu*, **Lu Zhang***. Yaobao: An evidence-based Chinese patent medicine assistant based on large language models and multi-source datasets. Under Review in **Phytomedicine (Q1, IF: 8.3)**
- Yu Xu, Bohao Zou, Guoxiang Xiu, Wei Jia*, **Lu Zhang***. Decoding heterogeneous aging clocks and disease risk stratification using a metabolomic foundation model. Under Review in **Nature Communications (Q1, IF: 14.7)**

2026

- Honghao Lai, Bernardo Sousa-Pinto, Christian Cao, David Moher, Janne Estill, Jiayi Liu, Weilong Zhao, Yutong Wang, Ziyang Ye, Bo Tong, Zhenhua Yang, Xufei Luo, Bingyi Wang, Yimeng Li, Bei Pan, **Lu Zhang**, Jinhui Tian, Yaolong Chen, Nannan Shi, Long Ge. Large language models for systematic reviews were reported to perform well but rarely with verifiable safeguards: a cross-sectional study. **Journal of Clinical Epidemiology (Q1, IF: 5.8)**. doi: 10.1016/j.jclinepi.2026.112383
- Shujun Xu, Junfang Lyu, Gengyu Bao, Ping Zheng, Yaqi Li, Ziwan Ning, Baohua Li, Min Zhuang, Hoi Ki Wong, Mingchun Wang, Yufei Zeng, Ya Zheng, Jingyuan Luo, Qin Liu, **Lu Zhang**, Chengyuan Lin, Haitao Xiao, Chun-Su Yuan, Hoi Leong Xavier Wong, Zhaoxiang Bian, Lixiang Zhai. Gut bacteria that produce fatty acid ethanolamides alleviate diarrhea-predominant IBS with insulin resistance. **Cell Host & Microbe (Q1, IF: 23.2)**. doi: 10.1016/j.chom.2026.05.020
- Keke Ding, Ruimin Xu, Xiaowen Chao, Mengci Li, Tao Sun, Tianbiao Yang, Min Zhu, Xufei Peng, Ruofan Liu, Peiyang Luo, Guoli Tian, Yun Lin, Guoxiang Xie, Xiaojiao Zheng, **Lu Zhang**, Wei Jia, Tianlu Chen. Amino acid-based biological age clock and its implications for human health and aging. **Nature Communications (Q1, IF: 14.7)**. doi: 10.1038/s41467-026-73371-y
- Yang Liu, Kexin Ma, Haoran Xu, Ke Xu, Yunfei Hu, Zhenhan Lin, Jiangli Lin, Bo Han, Shuaicheng Li, Zhixiang Lin, Xin Maizie Zhou*, **Lu Zhang***. Interpretable spatial multi-omics data integration and dimension reduction with SpaMV. **Nature Communications (Q1, IF: 14.7)** doi:10.1038/s41467-026-74718-1.
- Chonghao Wang, Werner Pieter Veldsman, **Lu Zhang***. Detection of short identity by descent segments using low-frequency variants. **IEEE/ACM Transactions on Computational Biology and Bioinformatics (Q1, IF:3.6)**. doi: 10.1109/TCBBIO.2026.3678898
- Jingjing Wang[†], Zhenmiao Zhang[†], Yang Chen[†], Xi Zhou, Jiabin Xiang, Chao Yang, Dmitry A. Rodionov, Andrei L. Osterman, Qinwei Qiu, Yusheng Deng, Yanmin Liu, Chengrui Wang, Xiaoxiao Shang, Li Huang, Chen Sun, Jianwen Guo, Zhimin Yang, Lijuan Han, Lixiang Zhai, Zhaoxiang Bian, Lijuan Han*, Wei Jia*, Xiaodong Fang*, **Lu Zhang***. Exploring differences in the human gut microbiome between Han Chinese and non-Chinese

populations. **Genome Biology (Q1, IF:10.1)**. doi: 10.1186/s13059-026-03932-3

- Ke Xu, Xin Maizie Zhou*, **Lu Zhang***. stDyer-image improves clustering analysis of spatially resolved transcriptomics and proteomics with morphological images. **Bioinformatics (Q1, IF:5.8)**. doi: 10.1093/bioinformatics/btag071

2025

- Yunfei Hu, Zhenhan Lin, Manfei Xie, Weiman Yuan, Yikang Li, Mingxing Rao, Yichen Henry Liu, Wenjun Shen, **Lu Zhang***, Xin Maizie Zhou*. MaskGraphene: an advanced framework for interpretable joint representation for multi-slice, multi-condition spatial transcriptomics. **Genome Biology (Q1, IF:10.1)**. doi: 10.1186/s13059-025-03850-w
- Fengxian Chen, Yan Li, Yaolong Chen, Zhaoxiang Bian, La Duo, Qingguo Zhou*, **Lu Zhang***, ADVANCED Working Group. Strategies for the Analysis and Elimination of Hallucinations in Artificial Intelligence Generated Medical Knowledge. **Journal of Evidence-Based Medicine (Q1, IF: 3.5)** doi: 10.1111/jebm.70075
- Xufei Luo[†], Zeming Li[†], Zhenhua Yang[†], Yanfang Ma, Fengxian Chen, Qi Wang, Long Ge, James Zou*, **Lu Zhang***, Yaolong Chen*, Zhaoxiang Bian, on behalf of the ADVANCED working group. Using large language models to assess the consistency of randomized controlled trials on AI interventions with CONSORT-AI: a cross-sectional survey. **Journal of Medical Internet Research (Q1, IF:5.7)**. doi: 10.2196/72412
- Xufei Luo, Bingyi Wang, Yule Li, Shuang Liu, Haodong Li, Yaxuan Ren, Wah Yang, Kyle Lam, Stephen R Ali, Gemma Sharp, Fabio Ynoe Moraes, Ye Wang, Di Zhu, Zhenhua Yang, Daher Mohammad, Robert Fruscio, Maged N. Kamel Boulos, Zhicheng Lin, Kazuki Ide, Xuping Song, **Lu Zhang**, Yih Chung Tham, Hui Liu, Long Ge, Yaolong Chen*, Zhaoxiang Bian, the GAMER working group and ADVANCED working group. Knowledge and Awareness of Generative Artificial Intelligence Use in Medicine Among International Stakeholders: A Cross-Sectional Study. **Journal of Evidence-Based Medicine (Q1, IF: 3.5)** doi: 10.1111/jebm.70034
- Yu Xu, Chonghao Wang, Ke Xu, Yi Ding, Aiping Lyu*, **Lu Zhang***. TRAFICA: An Open Chromatin Language Model to Improve Transcription Factor Binding Affinity Prediction. **Bioinformatics (Q1, IF:5.8)** doi: 10.1093/bioinformatics/btaf469
- Xufei Luo, Bingyi Wang, Qianling Shi, Zijun Wang, Honghao Lai, Hui Liu, Yishan Qin, Fengxian Chen, Xuping Song, Long Ge, **Lu Zhang**, Zhaoxiang Bian, Yaolong Chen*, ADVANCED working group. Lack of methodological rigor and limited coverage of generative artificial intelligence in existing artificial intelligence reporting guidelines: a scoping review. **Journal of Clinical Epidemiology (Q1, IF:5.2)**. doi: 10.1016/j.jclinepi.2025.111903
- Zirui Wang, Ke Xu, Yang Liu, Yu Xu, **Lu Zhang***. Mitigation of multi-scale biases in cell-type deconvolution for spatially resolved transcriptomics using HarMoDecon. **Bioinformatics (Q1, IF:5.8)**. doi: 10.1093/bioinformatics/btaf451
- Zeming Li, Yu Xu, Debajyoti Chowdhury, Hip Fung Yip, Chonghao Wang, **Lu Zhang***. Causal Transformer for Learning Embeddings from Structured Medical History Records and Multi-Source Data Integration for Complex Disease Risk Prediction. **Interdisciplinary Sciences: Computational Life Sciences (Q1, IF:4.39)**. doi.org/10.1007/s12539-025-00749-9
- Xiangzhe Yuan, Chonghao Wang, **Lu Zhang***. DBNX: A Machine Learning Method for Ensembling Polygenic Risk Scores and Non-Genetic Factors. **IEEE/ACM Transactions on Computational Biology and Bioinformatics (Q1, IF:3.6)**. doi: 10.1109/TCBBIO.2025.3580193
- Hiufung Yip, Zeming Li, **Lu Zhang***, Aiping Lyu*. Large Language Models in Integrative Medicine: A Review of Progress, Challenges, and Opportunities. **Journal of Evidence-Based Medicine (Q1, IF: 3.5)**. doi: 10.1111/jebm.70031
- Zeming Li[†], Xufei Luo[†], Zhenhua Yang[†], Huayu Zhang, Long Ge, Zhaoxiang Bian, James

- Zou*, Yaolong Chen*, **Lu Zhang***, on behalf of the ADVANCED working group. RAPID: Reliable and efficient automatic generation of submission reporting checklists with large language models. **Journal of the American Medical Informatics Association (Q1, IF: 4.7)** doi: 10.1093/jamia/ocaf093
- Ke Xu, Yu Xu, Zirui Wang, Xin Maizie Zhou*, **Lu Zhang***. stDyer enables spatial domain clustering with dynamic graph embedding. **Genome Biology (Q1, IF:10.1)**. doi: 10.1101/2024.05.08.593252
 - Honghao Lai, Jiayi Liu, Chunyang Bai, Hui Liu, Bei Pan, Xufei Luo, Liangying Hou, Weilong Zhao, Danni Xia, Jinhui Tian, Yaolong Chen, **Lu Zhang**, Janne Estill, Jie Liu, Xing Liao, Nannan Shi, Xin Sun, Hongcai Shang, Zhaoxiang Bian, Kehu Yang, Luqi Huang & Long Ge On behalf of ADVANCED Working Group. Language models for data extraction and risk of bias assessment in complementary medicine. **npj Digital Medicine (Q1, IF: 12.4)** doi: 10.1038/s41746-025-01457-w
 - Xin Yang, Chi Ho Chan, Shanshan Yao, Hang Yin Chu, Minchuan Lyu, Ziqi Chen, Huan Xiao, Yuan Ma, Sifan Yu, Fangfei Li, Jin Liu, Luyao Wang, Zongkang Zhang, Bao-Ting Zhang, **Lu Zhang**, Aiping Lu, Yaofeng Wang, Ge Zhang, Yuanyuan Yu. DeepAptamer: Advancing high-affinity aptamer discovery with a hybrid deep learning model. **Molecular Therapy-Nucleic Acids (Q1, IF:6.5)**. doi: 10.1016/j.omtn.2024.102436
 - Juan Shen, Weiming Liang, Ruizhen Zhao, Yang Chen, Yaming Liu, Wei Cheng, Tailiang Chai, Yin Zhang, Silian Chen, Jiazhe Liu, Xueting Chen, Yusheng Deng, Zhao Zhang, Yufen Huang, Huanjie Yang, Li Pang, Qinwei Qiu, Haohao Deng, Shanshan Pan, Linying Wang, Jingjing Ye, Wen Luo, Xuanting Jiang, Xiao Huang, Wanshun Li, Lixian Liang, **Lu Zhang**, Li Huang, Zhimin Yang, Rouxi Chen, Junpu Mei, Zhen Yue, Hong Wei, Karsten Kristiansen, Lijuan Han, Xiaodong Fang. Cross-tissue multi-omics analyses reveal the gut microbiota's absence impacts organ morphology, immune homeostasis, bile acid, and lipid metabolism. **iMeta (Q1, IF: 33.2)**. doi: 10.1002/imt2.272
 - Werner Pieter Veldsman, Qi Zhang, Qian Zhao, **Lu Zhang***. The antimicrobial peptides pipeline: a bacteria-centric AMP predictor. **Current Gene Therapy (Q2, IF: 3.8)**. doi: 10.1101/2024.05.26.595993
 - Bingyi Wang, Xufei Luo, Jie Zhang, Qianling Shi, Honghao Lai, Hui Liu, Yuanyuan Yao, Long Ge, Jiayi Liu, Haodong Li, Yanfang Ma, **Lu Zhang**, Zhaoxiang Bian, MD, Ivan D. Florez, Yaolong Chen, Janne Estill. Evaluating the Quality of Guidelines Using the AGREE II Tool by a Large Language Model vs Human Appraisers. **JAMA Network Open (Q1, IF: 10.5)**. doi:10.1001/jamanetworkopen.2025.7199
 - Xufei Luo, Long Ge, **Lu Zhang**, Yaolong Chen, Liang Du, ADVANCED working group. AI-Empowered Evidence-Based Research and Clinical Decision-Making. **Journal of Evidence-Based Medicine (Q1, IF: 3.5)**. doi: 10.1111/jebm.70014

2024

- Shaomin Zou, Chao Yang, Jieping Zhang, Dan Zhong, Manqi Meng, **Lu Zhang**, Honglei Chen* and Lekun Fang*. Multi-omic profiling reveals associations between the gut microbiome, host genome and transcriptome in patients with colorectal cancer. **Journal of Translational Medicine (Q1, IF:7.5)**. doi: 10.1186/s12967-024-04984-4
- Xu Wang, Hui Liu, Janne Estill, Fujian Song, Akihiko Ozaki, Juan V. A. Franco, Ivan D. Florez, Etienne Ngeh, Nav Persaud, Liliya Eugenevna Ziganshina, Myeong Soo Lee, **Lu Zhang**, Yuan Chi, Yuting Duan, Enmei Liu, Yaolong Chen*, Xiaodong Zhao*. Fostering innovation and advancement in evidence-based practice and guidelines within the field of pediatrics: The 2024 BRIGHT declaration. **Pediatric Discovery**. doi: 10.1002/pdi3.2513
- Yaxuan Ren, Xufei Luo, Ye Wang, Haodong Li, Hairong Zhang, Zeming Li, Honghao Lai, Xuanlin Li, Long Ge, Janne ESTILL, **Lu Zhang**, Shu Yang*, Yaolong Chen*, Chengping Wen, Zhaoxiang Bian, ADVANCED Working Group. Large Language Models in Traditional Chinese Medicine: A Scoping Review. **Journal of Evidence-Based Medicine (Q2, IF:3.6)**. doi: 10.1111/jebm.12658

- Bohao Zou, JingJing Wang, Yi Ding, Zhenmiao Zhang, Yufen Huang, Xiaodong Fang, Ka Chun Cheung, Simon See and **Lu Zhang***. A multi-modal deep language model for contaminant removal from metagenome-assembled genomes. **Nature Machine Intelligence (Q1, IF:18.8)**. doi: 10.1038/s42256-024-00908-5
- Xufei Luo, Han Lyu, Zaiwei Song, Hui Liu, Zhixiang Wang, Haodong Li, Ye Wang, Di Zhu, **Lu Zhang***, Yaolong Chen*. The Impact of Generative Artificial Intelligence on the Development, Evaluation, and Application of Clinical Practice Guidelines. **Medical Journal of Peking Union Medical College Hospital**. doi: 10.12290/xhyxzz.2024-0602
- Chao Yang, Zhenmiao Zhang, Yufen Huang, Xuefeng Xie, Herui Liao, Jin Xiao, Werner Pieter Veldsman, Kejing Yin, Xiaodong Fang*, **Lu Zhang***. LRTK: A platform agnostic toolkit for linked-read analysis of both human genomes and metagenomes. **GigaScience (Q1, IF: 11.8)**. doi: 10.1093/gigascience/giae028
- Zhenmiao Zhang, Jin Xiao, Hongbo Wang, Chao Yang, Yufeng Huang, Zhen Yue, Yang Chen, Lijuan Han, Kejing Yin, Aiping Lyu, Xiaodong Fang, **Lu Zhang***. Exploring high-quality microbial genomes by assembling short-reads with long-range connectivity. **Nature Communications (Q1, IF: 14.7)**. doi: 10.1038/s41467-024-49060-z
- Werner Pieter Veldsman, Chao Yang, Zhenmiao Zhang, Yufen Huang, Debajyoti Chowdhury and **Lu Zhang***. Structural and Functional Disparities within the Human Gut Virome in terms of Genome Topology and Representative Genome Selection. **Viruses-Basel (Q1, IF: 3.8)**. doi:10.3390/v16010134

2023

- Hiu Fung Yip, Xiao Wei, Zeming Li, Qing Ren, DongSheng Cao, **Lu Zhang***, Aiping Lu. Bio-Mol:Pretraining Multimodality Bioactivity Profile for Enhancing Small Molecule Property Prediction. bioRxiv. doi: 10.1101/2023.11.02.565401
- Zhenmiao Zhang, Mingxing Rao, Yufen Huang, **Lu Zhang***. A graph-based Gaussian Mixture Variational Autoencoder improves metagenome binning for short contigs. bioRxiv. doi: 10.21203/rs.3.rs-3308172/v1
- Zhenmiao Zhang, Chao Yang, Werner Pieter Veldsman, Xiaodong Fang, **Lu Zhang***. Benchmarking genome assembly methods on metagenomic sequencing data. **Briefings in Bioinformatics (Q1, IF: 6.8)**. doi: 10.1093/bib/bbad087
- Jingjing Wang, Werner Pieter Veldsman, Xiaodong Fang, Yufen Huang, Xuefeng Xie, Aiping Lyu, **Lu Zhang***. Benchmarking multi-platform sequencing technologies for human genome assembly. **Briefings in Bioinformatics (Q1, IF: 6.8)**. doi: 10.1093/bib/bbad3
- Ke Xu, ChinWang Cheong, Werner Pieter Veldsman, Aiping Lyu*, William K. Cheung*, **Lu Zhang***. Accurate and interpretable gene expression imputation on scRNA-seq data using IGSimpute. **Briefings in Bioinformatics (Q1, IF: 6.8)**. doi: 10.1093/bib/bbad124
- Lixiang Zhai, Haitao Xiao, Chengyuan Lin, Yan Y Lam, Hoi Leong Xavier Wong, Mengxue Gong, Guojun Wu, Yusheng Deng, Ziwan Ning, Chunhua Huang, Yijing Zhang, Min Zhuang, Chao Yang, **Lu Zhang**, Ling Zhao, Chenhong Zhang, Xiaodong Fang, Wei Jia, Liping Zhao, Zhao-xiang Bian. Gut microbiota-derived tryptamine impairs insulin sensitivity. **Nature Communications (Q1, IF: 14.7)**. doi: 10.1038/s41467-023-40552-y
- Werner P. Veldsman, Xin Maizie Zhou, Yuanwei Zhang, Bailiang Li, **Lu Zhang***. Editorial: Predicting high-risk individuals for common diseases using multi-omics and epidemiological data, volume II. **Frontiers in Genetics (Q2, IF: 2.8)**. doi: 10.3389/fgene.2023.1280648

2022

- Lixiang Zhai, Chunhua Huang, Ziwan Ning, Yijing Zhang, Min Zhuang, Wei Yang, Xiaolei Wang, Jingjing Wang, **Lu Zhang**, Haitao Xiao, Ling Zhao, Yan Y Lam, Chi Fung Willis Chow, Jiandong Huang, Shuofeng Yuan, Kui Ming Chan, Hoi Leong Xavier Wong, Zhao-xiang Bian. Ruminococcus gnavus plays a pathogenic role in diarrhea-predominant irritable

bowel syndrome by increasing serotonin biosynthesis. **Cell Host and Microbe** (Q1, IF: 31.3). doi: 10.1016/j.chom.2022.11.006

- Chonghao Wang, Jing Zhang, Xin Zhou, **Lu Zhang***. A comprehensive investigation of the statistical and machine learning approaches for human complex disease prediction. **Briefings in Bioinformatics** (Q1, IF: 6.8). doi: 10.1093/bib/bbac552
- Yu Xu, Jiaying Chen, Aiping Lyu, William K Cheung, **Lu Zhang***. dynDeepDRIM: a dynamic deep learning model to infer direct regulatory interactions using single cell time-course gene expression data. **Briefings in Bioinformatics** (Q1, IF: 6.8). doi: 10.1093/bib/bbac424
- Feng Shi, Delbert Almerick T Boncanc, Hin Ting Wan, Ting Fung Chan, **Lu Zhang**, Keng Po Lai, Chris Kong-Chu Wong. Hepatic metabolism gene expression and gut microbes in offspring, subjected to in-utero PFOS exposure and postnatal diet challenges. **Chemosphere** (Q1, IF: 8.1). doi: 10.1016/j.chemosphere.2022.136196.
- Hongbo Wang, Hin Ting Wan, Bin Wu, Jianbo Jian, Alice H M Ng, Claire Yik-Lok Chung, Eugene Yui-Ching Chow, Jizhou Zhang, Anderson O L Wong, Keng Po Lai, Ting Fung Chan, **Lu Zhang***, Chris Kong-Chu Wong*. A Chromosome-level Assembly of the Japanese Eel Genome, Insights into Gene Duplication and Chromosomal Reorganization. **GigaScience** (Q1, IF: 11.8). doi: 10.1093/gigascience/giac120
- Yunfei Hu, Sanidhya Mangal, **Lu Zhang**, Xin Zhou. Automated filtering of genome-wide large deletions through an ensemble deep learning framework. **Methods** (Q1, IF: 4.2). doi:10.1016/j.ymeth.2022.08.001

2021

- Chao Yang, Debajyoti Chowdhury, Zhenmiao Zhang, William K. Cheung, Aiping Lu, ZhaoXiang Bian, **Lu Zhang***. A review of computational tools for generating metagenome-assembled genomes from metagenomic sequencing data. **Computational and Structural Biotechnology Journal** (Q1, IF: 4.4). doi: 10.1016/j.csbj.2021.11.028
- Zi-Hang Wen, Jeremy L. Langsam, **Lu Zhang**, Wenjun Shen, Xin Zhou. A Bayesian factorization method to recover single-cell RNA sequencing data. **Cell Reports Methods** (Q2, IF: 4.3). doi: 10.2139/ssrn.3867727
- Li Song, **Lu Zhang***, Xiaodong Fang*. Characterizing enterotypes in human metagenomics: a viral perspective. **Frontiers in Microbiology** (Q2, IF: 4). doi: 10.3389/fmicb.2021.740990
- Yichen Henry Liu, Griffin L. Grubbs, **Lu Zhang**, Xiaodong Fang, David L. Dill, Arend Sidow, Xin Zhou. Aquila_stLFR: diploid genome assembly based structural variant calling package for stLFR linked-read. **Bioinformatics Advances** (Q4, IF: 2.4). doi:10.1093/bioadv/vbab007/6300508.
- Debajyoti Chowdhury, Maizie (Xin) Zhou, Bailiang Li, Yuanwei Zhang, William K Cheung, Aiping Lu, **Lu Zhang***. Predicting High-Risk Individuals for Common Diseases Using Multi-Omics and Epidemiological Data. **Frontiers in Genetics** (Q2, IF: 2.8). doi: 10.3389/fgene.2021.737598.
- Jiaying Chen, Chinwang Cheong, Liang Lan, Xin Zhou, Jiming Liu, Aiping Lyu, William K Cheung*, **Lu Zhang***. DeepDRIM: a deep neural network to reconstruct cell-type-specific gene regulatory network using single-cell RNA-Seq Data. **Briefings in Bioinformatics** (Q1, IF: 6.8). doi: 10.1093/bib/bbab325.
- Md Selim Reza, Yunpeng Cai, **Lu Zhang**, Xingyu Zhang, Yanjie Wei. Computational Solutions for Microbiome and Metagenomics Sequencing Analyses. **Frontiers in Molecular Biosciences** (Q1, IF: 3.9). doi: 10.3389/fmolb.2021.698384.
- Zhenmiao Zhang, **Lu Zhang***. METAMVGL: a multi-view graph-based metagenomic contig binning algorithm by integrating assembly and paired-end graphs. **BMC Bioinformatics** (Q1, IF: 2.9). doi: 10.1186/s12859-021-04284-4.
- Xin Zhou, **Lu Zhang**, Ziming Weng, David L. Dill, Arend Sidow. Aquila enables reference-assisted diploid personal genome assembly and comprehensive variant detection

based on linked reads. **Nature Communications** (Q1, IF: 14.7). doi: 10.1038/s41467-021-21395-x

2020

- **Lu Zhang***, Xiaodong Fang et al. A Comprehensive Investigation of Metagenome Assembly by Linked-Read Sequencing. **Microbiome** (Q1, IF: 13.8). doi: 10.1186/s40168-020-00929-3.
- **Lu Zhang**[†], Xin Zhou[†], Ziming Weng, Arend Sidow. De novo diploid genome assembly for genome-wide structural variant detection. **NAR Genomics and Bioinformatics** (Q1, IF: 4). doi: 10.1093/nargab/lqz018

2019

- **Lu Zhang**[†], Xin Zhou[†], Ziming Weng, Arend Sidow. Assessment of human diploid genome assembly with 10x Linked-Reads data. **GigaScience** (Q1, IF: 11.8). doi: 10.1093/gigascience/giz141

2018

- JiFeng Guo[†], **Lu Zhang**[†] et al. De novo coding mutations contribute to early onset Parkinson's disease. **Proceedings of the National Academy of Sciences of the United States of America** (Q1, IF: 9.4). doi: 10.1073/pnas.1809969115.
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Teaching experience

2026 Spring **COMP3115** Exploratory Data Analysis and Visualization
 2025 Fall **COMP7990** Principles and Practices of Data Analytics
 2025 Spring **COMP3115** Exploratory Data Analysis and Visualization
 2024 Fall **COMP7990** Principles and Practices of Data Analytics
 2024 Spring **COMP3115** Exploratory Data Analysis and Visualization
 2023 Fall **COMP1007** Introduction to Python and Its Applications
 2023 Fall **COMP7990** Principles and Practices of Data Analytics
 2023 Spring **COMP1007** Introduction to Python and Its Applications
 2023 Spring **COMP3115** Exploratory Data Analysis and Visualization
 2022 Fall **COMP7990** Principles and Practices of Data Analytics
 2022 Spring **COMP1007** Introduction to Python and Its Applications
 2022 Spring **COMP3115** Exploratory Data Analysis and Visualization
 2021 Fall **COMP7990** Principles and Practices of Data Analytics
 2020 Fall **COMP1007** Introduction to Python and Its Applications
 2020 Fall **COMP7990** Principles and Practices of Data Analytics
 2019 Fall **COMP1007** Introduction to Python and Its Applications
 2019 Fall **COMP7990** Principles and Practices of Data Analytics
 2015 Fall (TA) **GE2334** The Art and Science of Data
 2014 Spring (TA) **CS2312** Problem Solving and Programming
 2013 Fall (TA) **CS4335** Design and Analysis of Algorithms
 2013 Spring (TA) **CS2312** Problem Solving and Programming

Service

- Associate Program Director of BSc (Hons) in Computer Science (2025-now)
- Department UoA coordinator of Research Assessment Exercise 2026
- Program management committee member of BSc (Hons) in computer science (2022-now)
- Program management committee member of MSc in Data Analytics and Artificial Intelligence (2022-now)
- Program management committee member of MSc in Information Technology Management (2022-now)
- Program management committee member of BSc (Hons) in Innovation in Health/Well Being (2022-2024)
- Department Coordinator of Computer Facility (2023-2025)

Conference organization and committee members

- NeurIPS workshop reviewer (2025,2026)
- Area Chair ACM-BCB 2025
- PC members of GIW2023, BIBM (2020-2026), ISBRA (2020-2026), ISDH 2019, ICBPB 2019, IJCAI (2020, 2021), BDAI 2020, ICHI (2020, 2021), ICBBS 2024, GIW (2025, 2026)
- Organization committee member, 2019 International Symposium on Digital Health
- Organization Chair of “Computational methods and their applications on single cell multiomic data” workshop in BIBM2021
- Organization Chair of “Computational methods to characterize genomic variants using high-throughput sequencing data” workshop in BIBM2022
- Organization Chair of “Algorithmic Advances for Single-Cell and Spatial Omics Data Analysis” workshop in BIBM2025
- Organization Chair of “Algorithmic Advances for Single-Cell and Spatial Omics Data Analysis” workshop in BIBM2026
- AI leader and co-founder of ADVANCED (AI-empowered Development and Assessment of medical guidelines and standards) working group (<http://www.advancedguides.ai/>)
- Member of international RIGHT (Reporting Items for Practice Guidelines in Healthcare) working group
- Member of PREPARE (Practice Guideline Registration and Transparency) platform working group
- Member of STAR (Scientific, Transparent and Applicable Rankings tool) working group

Editorial members

- Editor-In-Chief AI Evidence-Based Medicine
- Editorial Board member, BMC Methods.
- Editorial Board member, Current Gene Therapy.
- Guest Editor, Life, Advances in Computational Genomics, 2022.
- Guest Editor, Frontiers in Genetics, Predicting High-Risk Individuals for Common Diseases Using Multi-Omics and Epidemiological Data I, 2019
- Guest Editor, Frontiers in Molecular Biosciences, Computational Solutions for Microbiome and Metagenomics Sequencing Analyses I, 2020
- Guest Editor, Frontiers in Genetics, Predicting High-Risk Individuals for Common Diseases Using Multi-Omics and Epidemiological Data II, 2022
- Guest Editor, Frontiers in Molecular Biosciences, Computational Solutions for Microbiome and Metagenomics Sequencing Analyses II, 2021
- Guest Editor, Health Information Science and Systems, Integrative multi-omics for human health and disease studies, 2022

Journal Reviewers

Nature Methods, Nature Communications, Genomics, Proteomics & Bioinformatics, Genome Biology, Genome Research, Genome Medicine, GigaScience, Bioinformatics, iScience, Briefings in Bioinformatics, npj Precision Oncology, IEEE/ACM Transactions on Computational Biology

and Bioinformatics, National Science Review, Frontiers in Genetics, Knowledge-Based Systems, Aging, Computational Biology and Chemistry et al.